

An Overview of Robot Calibration

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Abstract—An overview is given of the existing work on robot calibration, and some of the basic issues are identified in calibration and improvement of robot precision. Modeling, measurement, identification, and correction issues in robot calibration are discussed, and some of the unresolved questions are identified.

INTRODUCTION

CALIBRATION techniques can be used to improve robot accuracy through software rather than changing the mechanical structure or design of the robot. Calibration can also minimize the risk of having to change application programs due to slight changes or drifts (such as wear of parts, dimensional drifts and tolerances, and component replacement effects) in the robot system. This is especially important in applications that may involve a rather large number of task points.

Our goal in this paper is to provide an overview of the subject and to identify some of the related research issues. No new research results or mathematical developments are presented in the paper.

For the purpose of reviewing the existing work, robot calibration is classified into three levels. At each level modeling, measurement, identification, and correction issues are discussed, and the relevant research work is reviewed.

ROBOT CALIBRATION

Robot calibration is a process by which robot accuracy can be improved by modifying the robot positioning software rather than changing or altering the design of the robot or its control system. It involves identifying a more accurate functional relationship between the joint transducer readings and the actual workspace position of the end effector and using these identified changes to permanently change (between each consecutive calibration) the robot positioning software. Calibration is a discrete event and as such is different from adaptive control (of robots) where model identification is carried continuously and controller parameters are adjusted in accordance with the identified changes. The foregoing definition of calibration assumes that a nominal relationship between the end-effector position and joint transducer readings is

known but that this relationship is not necessarily accurate due to changes in robot hardware or uncertainties in the parameters of the nominal relationship.

Calibration procedures vary widely in their complexity. For example, some robot calibration procedures only consider the joint transducer information while others may involve changes in the kinematic or dynamic model of the robot. In an effort to classify most of the current approaches to robot calibration, we have chosen to define three levels of robot calibration.

For the purpose of this work, level 1 calibration shall be defined as "joint level" calibration. The goal is to determine the correct relationship between the signal produced by the joint displacement transducer and the actual joint displacement. This usually involves calibration of the kinematics of the drive and the joint sensor mechanisms. A level 2 calibration is defined as the entire robot kinematic model calibration. At this level the purpose of the calibration is to determine the basic kinematic geometry of the robot as well as the correct joint-angle relationships. Level 3 calibration is defined as "non-kinematic" (nongeometric) calibration. Nonkinematic errors in positioning of the end effector of a robot are due to effects, such as joint compliance, friction, and clearance, as well as link compliance. Also, if the robot is under dynamic (rather than kinematic) control, then correction for changes in the dynamic model of the robot constitutes a level 3 calibration.

In general, the calibration process at any level consists of four steps. The first step would be to choose the form of a suitable functional relationship. We will refer to this step as the modeling step. The second step would be to collect some data from the actual robot that relate the input of the model to the output. This step will be termed the measurement step since it consists of the physical data collection process. The third step would be the mathematical process of using the data collected to identify the coefficients in the model. We will call this step the identification step. Note that an important part of this step is the determination of the expected error in the identified coefficients because of the error or noise in the measurement process. The final step would be the implementation of the new model in the position control software of the robot. This will be referred to as the correction step.

We will address each of the three levels of calibration individually. At each level the four steps in the calibration process will be described, and current research in the area is highlighted.

LEVEL 1 CALIBRATION

At this level the goal of the calibration is to ensure that the reading from a joint sensor yields the correct joint displacement. In many cases, this procedure is done as a part of the

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construction of the robot, and the user may only have to repeat the calibration if damage has occurred or if the joint has been disassembled for maintenance. In some cases, however, this type of calibration must be performed every time the robot is powered up. For example, if a robot has incremental encoders, the position of the robot is not known when the power is turned on, and the robot must be moved to a reference position to reset the encoder counters. This "homing" process is usually automated and is, therefore, transparent to the user, but it still represents a level 1 calibration. Also, in many cases the automated "start-up" calibration relies on physical stops or flags that may shift. If this occurs, some additional calibration may be required to correct the normally automated procedure.

Model

The model that is appropriate at this level (level 1) is simply the relationship between the output of the joint sensor and the actual joint displacement. In the case that the position transducers are on the motor shaft, this model includes the kinematics of the drive system. If we define the signal from the transducer as η_i and the actual joint displacement as θ_i , the following relationship holds:

$$\theta_i = h_i(\eta_i, \gamma_i) \quad (1)$$

where $h_i(\)$ is the appropriate input-output functional relationship in explicit form and the vector γ_i represents the vector of parameters in the function $h(\)$. In the large majority of cases the function $h(\)$ is assumed to be linear and can be written as

$$\theta_i = k_{i1}\eta_i + k_{i2}. \quad (2)$$

In this model the vector γ_i will be $[k_{i1}, k_{i2}]^T$. The purpose of the level 1 calibration, therefore, would be to determine the values of the vector γ_i correctly.

In cases involving high precision it may become necessary to develop a more sophisticated model to describe the relationship between the transducer signal and the joint displacement. Nonlinear effects, such as gear backlash or joint clearance, for example, may be included in the model. It should be pointed out, however, that such effects are usually modeled nonkinematically and are therefore considered (in this paper) as level 3 calibration.

Measurement

For level 1 calibration the measurement process consists of either using some external measurement device to determine the actual joint angle accurately or moving the joint to some "known" configuration. The simplest technique for moving to a "known" joint configuration is to pick a configuration that may be easily verified through the robot geometry. For example, it may be known that a joint will be at a certain angle when two surfaces on the robot line up (become congruent) or when a link is "straight up." This technique is used, for example, in calibrating the PUMA 560 robot [1].

If a visual alignment is not possible or does not provide the required level of repeatability, the next approach is to use some type of fixture to ensure that a joint is at a known angle. This may be achieved by including alignment holes in the joint

design through which a close tolerance peg may be inserted or by using an external fixture that is attached to the robot structure.

Another approach involves placing the gripper in a known position (and orientation) in the workspace. The robot configuration that is necessary to achieve this position may be determined, and, therefore, all of the joint angles are specified. This known position in the workspace is often defined by a fixture. Of course, a measurement problem is associated with locating the fixture in the workspace. Measurement techniques for accomplishing this will be discussed with the level 2 calibration.

A different approach involves the use of external instrumentation to determine the joint angles accurately. For example, Whitney *et al.* [2] describe a system which employs a polygonal mirror and a theodolite. With this approach the operator looks through the theodolite and the joint is rotated until the operator sees his own reflection centered in the theodolite. When this occurs, the mirror surface is almost perfectly normal to the axis of the theodolite. The robot joint is then rotated to the next mirror segment and lined up again. Since the angle between the mirror segments is known very precisely, the joint rotation is also known. The particular measurement technique chosen at this level is a trade-off between precision and cost.

Identification

In the case of the linear model the identification process for level 1 calibration is trivial. The parameter k_{i1} is a combination of the gain of the transducer itself and ratio between the joint motion and the transducer motion. The transducer gain is usually specified very accurately by the manufacturer. The ratio between the transducer motion and joint motion is easily determined by examining the joint design. This ratio is usually established through a gear train or a similar device.

Once the gain term is determined, the remaining constant k_{i2} may be determined by placing the joint at a known displacement and then reading the transducer signal. The value of k_{i2} is simply the difference between the joint displacement θ_i and $k_{i1}\eta_i$.

If a nonlinear model has been selected, the identification procedure becomes slightly more involved. In this case some of the methods discussed in level 2 calibration may be appropriate.

Correction

The correction phase of level 1 calibration is very simple. In the vast majority of robots the signal that comes from the joint transducer is converted by the controller into a representation of the actual joint angle. This conversion may be in software, in the case of a digital computer, or the conversion may be accomplished through specialized circuitry. In either case the parameters that have been identified are represented in the conversion process. The process of correction is simply ensuring that the representations of the parameters are correct. In the case of software this implies storing the proper set of constants. For a hardware implementation a set of gains may be adjusted. In either case the correction process is straightforward.

LEVEL 2 CALIBRATION

The goal of a level 2 calibration is to improve the accuracy of the kinematic model of the manipulator as well as the relationship between the joint transducers and the actual joint displacement. This means that level 2 calibration also includes a level 1 calibration. For a level 2 calibration the links of the robot will be assumed to be rigid, and the joints of the robot will be assumed to be perfect in the sense that they would allow no undesired motion about their axes. What this means is that compensation for the errors due to joint clearances, for example, are not included as part of level 2 calibration. The objective of the calibration at level 2 will be to determine the spatial kinematic relationship between the joints and links.

If we define the vector x to be the 6-vector that describes the position (position of a point and orientation) of the end effector in space, then under the assumptions of a level 2 calibration x will be given by

$$x = g(\eta, \gamma, a) \quad (3)$$

Where η is the vector of joint transducer readings, γ is the vector of coefficients in the relationships between the joint transducers and the actual joint displacements, and a is the vector of coefficients in the kinematic model that is being used. For a level 2 calibration, therefore, the form of the kinematic model $g(\)$ must be chosen, and values for the vectors η , γ , and a must be determined.

Model

A number of different approaches exist for developing the kinematic model of a robot manipulator. The most popular method has been the procedure established by Denavit and Hartenberg [3] which is based on homogeneous transformation matrices. This procedure consists of establishing coordinate systems on each joint axis. Each coordinate system is then related to the next through a specific set of coefficients in the homogeneous transformation matrices. This technique is used by Wu [4] to examine robots with small variations in their kinematic arrangement. Unfortunately, a problem arises when two revolute joint axes are parallel. One of the parameters in the Denavit-Hartenberg model is the length of the common normal between two consecutive joint axes. If two axes are parallel, an infinite number of common normals exist, all of the same length. The problem arises when one of the axes becomes slightly misaligned. In this case the length of the common normal can assume virtually any positive value. This variation may not be continuous as the axes become misaligned. The result is that the coefficients in the kinematic model do not vary proportionally with the degree of misalignment. Discontinuities or very large variations can exist in the coefficients for small degrees of misalignments, which create severe numerical difficulties in the identification process. For this reason a number of investigators have chosen to use a modified form of the Denavit-Hartenberg formulation for all axes except those which are parallel or nearly parallel. Parallel axes are treated as special cases with a different formulation that allows the parameters to vary in proportion to the degree of misalignment. Similar approaches have been used by Hsu and Everett [5] and Whitney *et al.* [2], who modified the basic

procedure described by Denavit and Hartenberg. Ibarra and Ferreira [6] utilize the Denavit-Hartenberg procedure and then modify each transformation matrix with a differential screw matrix for small misalignments. Mooring and Tang [7] used the general spatial rigid body displacement equation to develop a model to handle manipulators with variations in the kinematic model.

In level 2 calibration the main issues in modeling seem to be the adequacy and numerical stability of the representation. The adequacy of the representation is the ability of the model to characterize changes in the kinematics of the robot in terms of a finite set of parameters. Stable representation implies that small changes in the kinematics of the robot will result in similar small changes in the model of the robot. The question of what is the best representation for kinematics of a manipulator for calibration purposes is still an open research problem.

Measurement

The measurement phase of Level 2 calibration involves workspace sensing of position of the end effector or tool of the robot. In general, six parameters are necessary (but not necessarily sufficient) to completely specify the position of a rigid body. The sufficiency requirements are related to the uniqueness of the specification and depend on the exact nature of the six conditions used to specify the position of the body. Sensing the position of a body, therefore, requires measurements that would be equivalent to imposing six conditions on the position of the body. In calibration work, since the purpose of the measurement is to identify the manipulator kinematics, it is not necessary to make complete measurement of the end-effector position. Instead, several equivalent incompletely specified position measurements will also be sufficient. If both the manipulator and the drive system kinematics are to be identified, then more measurements are needed when compared to the identification of position of the end effector alone. For the purpose of this paper we shall only consider sensing systems that impose constraints on the following geometric elements of points, lines, planes, and spheres in the end effector or the workspace.

Constraining a point of the end-effector to remain on a given plane or sphere in the workspace corresponds to one condition. The physical realization of this condition is when a robot hand-held touch or tactile sensor (such as a linear variable differential transformer (LVDT)) is brought into contact with a flat plane or surface of a fixture in the workspace. Six such simultaneous conditions are needed to completely determine the position of the end effector. In robot calibration work, usually a fixture consisting of a cube or three spherical surfaces is used in the workspace, and the hand-held sensor is brought into contact six times each with different sides of the cube (or twice with each of three sides) or twice with the surface of each of the three spheres. This means that one condition is measured for six different positions of the end effector, or six incompletely specified positions of the end effector are measured. These six conditions on different positions of the end effector can be used to determine six kinematic parameters from (3).

The kinematic inversion of this situation is when the sensing element is fixed in the workspace and the cubical or spherical fixture is held by the robot hand. Kinematically, this means that a plane or a spherical surface in the end effector is to pass through a given point (the tip of the sensor) in the workspace. This again imposes only one condition on the position of the end effector, and six of such conditions are needed to determine the position of the end effector. This case is the basis of many end-position sensing devices used in robotics work (see, for example, Mooring and Pack [8], Ranky [9], Mathews and Hill [10]).

Sensing systems can also be designed by constraining the position of lines (rather than points, planes, or spheres) of the end effector or the workspace. One possibility is to use an optical sensor in the robot hand (for example, between two fingers) and bring the hand into a position at which the optical beam just gets broken due to contact with the surface of a fixture in the workspace. Such an implementation is used in the IBM 7565 robot in relation to referencing positions of objects in its workspace to the robot coordinates (see [11]). Kinematically, every time a line in the end effector becomes tangent to a fixture surface in the workspace, two conditions are imposed on the position of the end effector. Three such conditions are therefore necessary to impose six conditions on the end-effector position(s).

It should be pointed out that there are many other ways to implement sensing systems to satisfy the type of constraints discussed or other types of constraints. One issue of basic interest in this area is the enumeration of all possibilities for sensor-fixture systems that can be used in level 2 robot calibration. Another issue of interest is a comparative study of measurement accuracy and cost of such sensor-fixture systems.

Identification

The identification of the parameters in a robot kinematic model is a problem that has been addressed by a number of researchers (see, for example, [2], [4]–[7], and [12]–[15]) using a variety of models and identification algorithms.

Consider two similar robots, A and B. The identification problem is to estimate the model of robot B given the model of robot A and a set of measurements made on robot B. Robot A usually represents the “perfect” or nominal robot which, essentially, provides the functional structure $g(\cdot)$. The problem is thus reduced to a parameter identification problem. Specifically, in level 2 calibration the vectors γ_B and a_B are unknown. Estimates $\hat{\gamma}_B$ and $\hat{a}_B$ are to be constructed based on a set of measurement data. The end effector of robot B is placed at m locations $\{x_B(1), \dots, x_B(m)\}$ within the robot workspace. For each of the m positions the relationship between the workspace position and the joint displacement transducers will be given by

$$x_B(j) = g(\eta_B(j), \gamma_B, a_B), \quad j = 1, \dots, m \quad (4)$$

in which end-point sensing is used to determine some of the elements of each vector $x_B(j)$, $j = 1, \dots, m$. Let $y_B(j)$ be the subset of x_B that is measured or determined from the

constraint equations. Then

$$y_B(j) = g_y(\eta_B, \gamma_B, a_B), \quad j = 1, \dots, m \quad (5)$$

where g_y represents the appropriate subset of the kinematic equations.

The measurement equations are modeled as

$$y_B(j) = \hat{y}_B(j) + v_B(j), \quad j = 1, \dots, m \quad (6)$$

where $\hat{y}_B(j)$ are read from the end-point sensors or calculated using the end-point sensing data, and $v_B(j)$ is the measurement error (noise). In addition, at each robot configuration $x_B(j)$ the joint displacement sensors $\eta_B(j)$ are read. It is assumed that this reading is made without error. The parameter identification algorithm maps the measurement data

$$(\hat{y}_B(1), \eta_B(1)), \dots, (\hat{y}_B(m), \eta_B(m))$$

and the *a priori* data (i.e., nominal values of the kinematic parameters and possibly probabilistic data regarding the measurement noise and the unknown parameters) into a unique estimate of the vectors γ_B and a_B . In addition, the identification algorithm may provide for a measure of the estimation error.

The only restriction regarding the minimum number of measurements m is that enough data exist to provide a unique estimate. From a practical point of view, the answers to the following questions are presently unresolved research issues.

- 1) What is the relationship between the calibration error and robot repeatability?
- 2) What is the relationship between the calibration error and the accuracy of the calibration sensors?
- 3) How many measurements need to be taken to achieve a specified calibration accuracy?
- 4) What identification algorithm should be used?
- 5) Does additional modeling beyond the kinematic modeling (such as probabilistic characterization of measurement noise and unknown robot parameters) promise a significant improvement of the calibration accuracy?

To illustrate the importance of the foregoing questions, we further investigate the identification phase of level 2 calibration. Again, it should be pointed out that no new results are presented; only some of the relevant issues are discussed.

Starting with the ideal situation, $v_B(j) = 0$, $j = 1, \dots, m$ (zero measurement error), the identification problem reduces to the problem of solving nonlinear algebraic equations of the form

$$y_B(j) = \hat{y}_B(j) = g_y(\hat{\eta}_B(j), \hat{\gamma}_B, a_B), \quad j = 1, \dots, m. \quad (7)$$

To assure that the number of equations is not smaller than the number of unknowns, the minimum number of measurements m should satisfy the following inequality:

$$m \dim(y_B) \geq \rho + \zeta \quad (8)$$

where ρ is the number of elements in the vector γ and ζ is the number of elements in the vector a . Studies along these lines have been carried out by several researchers (see, for example, [4], [7], [14]). It should be pointed out that (8) is

only a necessary condition on the number of measurements—it does not account for observability issues. Such issues are discussed later in this section. In other words, it is assumed that the choice of the functional relationship $g_y(\cdot)$ is consistent with the task of identifying all the kinematic parameters.

Since the difference in the two robot models A and B may be expressed as

$$a_B = a_A + \Delta a \quad (9)$$

$$\gamma_B = \gamma_A + \Delta \gamma, \quad (10)$$

we may define the vector X to be

$$X = (\Delta \gamma^T, \Delta a^T)^T. \quad (11)$$

The relationship between X and the measurements $\hat{y}_B$ is of interest. We may define a “world coordinate error vector” $e(j)$ as follows:

$$e(j) = g_y(\eta_B(j), \gamma_B, a_B) - g_y(\eta_B(j), \gamma_A, a_A). \quad (12)$$

Many researchers have looked at (12) in one form or another. Some have used the formulation for robot calibration purposes [2], [5]–[7], [12]–[13], [15], and others have used it for the purpose of estimating robot accuracy (i.e., how errors at the joint level translate to the robot end effector) [4], [16]–[17]. The common thread in all of this work is the assumption that the world coordinate error vector $e(j)$ may be related to the vector of parameter offsets X through a linear transformation. This may be expressed as

$$e(j) = H(j)X. \quad (13)$$

We may define a “measurement vector” $z(j)$ as follows:

$$z(j) = \hat{y}_B(j) - g_y(\eta_B(j), \gamma_A, a_A) \quad (14)$$

where $z(j)$ is known since all of the parameters in (14) are either measured or defined by the nominal model A. From (6), $e(j)$ and $z(j)$ may be related as follows:

$$z(j) = H(j)X - v_B(j), \quad j = 1, \dots, m \quad (15)$$

where $H(j)$ is a matrix that depends on the nominal kinematic parameters and the robot configurations during the robot measurement phase. Equation (15) is the basis for the robot model identification procedure.

Let Z be the vector of all measurements:

$$Z = (z^T(1), \dots, z^T(m))^T. \quad (16)$$

Equation (15) may now be rewritten as

$$Z = HX + V \quad (17)$$

where

$$H = (H^T(1), \dots, H^T(m))^T \quad (18)$$

$$V = -(v_B^T(1), \dots, v_B^T(m))^T. \quad (19)$$

V and X are random vectors having certain probability distribution functions. The modeling of V and X is a

challenging task. Calibration measurement noise V depends on the accuracy and resolution of the end-point sensors, machining tolerances of the calibration fixtures, the method through which the sensor data are processed to provide the values of $\hat{y}_B(j)$ (see the section on level 2 measurement), as well as other factors. To properly model X , one should identify many possible sources of errors that may cause robot B to deviate from robot A. For example, one may wish to consider effects such as machining tolerances of certain robot links, axis misalignment, encoder mounting, and quantization noise, and many others. These are all examples of areas for further study.

The simplest approach to the identification problem is to ignore the models of both X and V . A unique least-squares estimate $\hat{X}$ is given by

$$\hat{X} = (H^T H)^{-1} H^T Z \quad (20)$$

under the conditions that $\dim(Z) \geq \dim(X)$ and $H^T H$ is nonsingular. Such an estimator has been used by a number of researchers [2], [18]. An interesting point raised by Wu [4] is that in many practical situations certain components of X may dominate others. For instance, errors in the Denavit–Hartenberg parameters of link-twist and joint-rotation angles may have a larger effect on robot accuracy than errors in the link-length parameters. These errors may also become more pronounced for joints closer to the robot base. This suggests the inclusion of an appropriate weighting matrix in the performance index.

We shall not turn attention to the probabilistic approach to identification. A plausible model for robot calibration is to assume that V and X are both Gaussian with zero mean and covariance matrices Σ_v and Σ_x , respectively. Furthermore, it is assumed that V and X are statistically independent of each other. Such models have been employed by Wu [4], Waldron and Kumar [17], and Hayati and Mirmirani [13]. It may be shown that a minimum-variance estimate $\hat{X}$ is given by

$$\hat{X} = (\Sigma_x^{-1} + H^T \Sigma_v^{-1} H)^{-1} H^T \Sigma_v^{-1} Z. \quad (21)$$

The error covariance $\Sigma_{\hat{X}} = E[(X - \hat{X})(X - \hat{X})^T]$ is given by

$$\Sigma_{\hat{X}} = (\Sigma_x^{-1} + H^T \Sigma_v^{-1} H)^{-1}. \quad (22)$$

This relates the calibration error to measurement noise, and the *a priori* kinematic parameter offsets uncertainty. Note that ignoring the model of X can be accommodated into (21) and (22) by setting $\Sigma_x^{-1} = 0$. A recursive representation of (21) and (22) is known as the Kalman filter. This approach to robot calibration has been taken by Whitney and Junkel [19] and Hayati and Mirmirani [13]. The Kalman filtering formulation is useful as it provides us with a framework to study the issues related to the number of measurements that need to be taken as well as the effects of robot repeatability. Consider the linear filtering problem:

$$X(j+1) = X(j), \quad \text{where } X(0) \sim N(0, \Sigma_x) \quad (23)$$

and

$$z(j) = H(j)X(j) + \tilde{v}(j), \quad \text{where } \tilde{v}(j) \sim N(0, \Sigma_v(j)) \quad (24)$$

where $N(-, -)$ denotes Gaussian distribution with the indicated mean and covariance and $\tilde{v}(j) = -v(j)$. The “process” equation (23) is that of a constant random process. We note that this system is time varying since $H(j)$, which represents the robot configurations, and $\Sigma_v(j)$, which represents the measurement noise, may vary from one measurement to the next. Under the assumption that $X(0)$ and $\tilde{v}(j)$ are independent, this problem is equivalent to that of (17) with the exception of leaving the total number of measurements m unspecified. We conjecture at this point that (23) may be generalized to include robot repeatability effects as follows:

$$X(j+1) = \Phi(j)X(j) + w(j) \quad (25)$$

where $W(j) \sim N(0, \Sigma_w(j))$.

Determining suitable “process dynamics” $\Phi(j)$ and “process noise” covariance $\Sigma_w(j)$ is a challenging research task. To determine how many measurements to take, the rate of convergence of the error covariance $\Sigma_x(j)$ needs to be studied.

We may now order the measurements $z(j)$ in the following fashion without any loss of generality. Let $K_p \geq K_{\min}$ be the number of measurement points and $K_r \geq 1$ be the number of repeated measurements at one measurement point. Then

$$m = K_p K_r. \quad (26)$$

The ordering of measurements that are fed to the filter is then as follows:

measurements taken at P_1 :

$$z(1), z(K_p + 1), \dots, z((K_r - 1)K_p + 1)$$

measurements taken at P_2 :

$$z(2), z(K_p + 2), \dots, z((K_r - 1)K_p + 2)$$

$\vdots$

measurements taken at P_{K_p} :

$$z(K_p), z(2K_p), \dots, z(K_r K_p).$$

A time-invariant version of the identification problem represented by (23) and (24) may be formulated by first defining

$$H_{K_p} = (H^T(1), \dots, H^T(K_p))^T \quad (27)$$

$$\Sigma_{v, K_p} = \text{diag} \{ \Sigma_v(j) \} \quad (28)$$

$$V_{K_p} = (v^T(1), \dots, v^T(K_p))^T. \quad (29)$$

Then

$$Z_{K_p} = H_{K_p} X(i) + V_{K_p}(i), \quad \text{where } i = 1, \dots, K_r \quad (30)$$

where $i = 1 + \text{int}((j-1)/K_p)$ and “int” represents the “largest integer not greater than.” For the system to be observable, we require that the rank $(H_{K_p}) = \dim(X)$, which implies that $K_{\min} \geq 1 + \text{int}(\dim(X)/\dim z(j))$. It should be pointed out that this time-invariant formulation does not include identification of the model of robot repeatability.

Using (22) repeatedly it may be shown that

$$\Sigma_x(i) = (\Sigma_x^{-1} + i H_{K_p}^T \Sigma_{v, K_p}^{-1} H_{K_p})^{-1}. \quad (31)$$

Hence if H_{K_p} is nonsingular and Σ_{v, K_p} is positive definite, then $\Sigma_x(i)$ goes to 0 as K_r goes to infinity. One may erroneously conclude that calibration can be made infinitely accurate as the number of measurement points is increased. Equation (31) is probably a manifestation of the practical inadequacy of the model in (23) and (24). Practice shows that a lower bound exists on the calibration error that is dictated by robot repeatability and calibration measurement error. This is captured by the modified model in (24) and (25). More research is required to clarify this aspect of the problem.

Correction

The two robots A and B, whose kinematic models $X = g(\eta, \lambda, a)$ are by now known such that model A is known perfectly and model B is known with a certain level of uncertainty, are assumed to share the same task. Typically, two types of tasks exist. *Data driven tasks* are described in terms of a sequence of k target world coordinate positions $\{x(1), \dots, x(k)\}$, and *taught tasks* are described in terms of a sequence of k joint level points $\{\eta(1), \dots, \eta(k)\}$. Ideally, one expects to find at each task point

$$\begin{aligned} x(j) &= x_A(j) = g(\eta_A(j), \gamma_A, a_A) \\ &= x_B(j) = g(\eta_B(j), \gamma_B, a_B), \quad j = 1, \dots, k. \end{aligned}$$

In practice, however, we hope to have

$$x(j) = x_A(j) = g(\eta_A(j), \gamma_A, a_A) \quad (32)$$

$$x_B(j) = g(\eta_B(j), \hat{\gamma}_B, \hat{a}_B) + \epsilon_{id}(j) \quad (33)$$

$$\|x(j) - x_B(j)\| \leq \epsilon_{ac}(j) \quad (34)$$

where $\epsilon_{ac}(j)$ is a prescribed accuracy measure attached to every task point and ϵ_{id} represents world coordinate errors due to imperfect calibration. The correction phase is the process of implementing the identified model of robot B to satisfy the requirement expressed in (34). In the case of a data-driven application, the model correction phase for robot B can be implemented (at least conceptually) in the following manner.

Step 1: Substitute $x_B(j) = x(j)$ (i.e., ignoring the unknown errors, $\epsilon_{id}(j)$).

Step 2: From the inverse kinematics of $x(j) = g(\eta_B(j), \hat{\gamma}_B, \hat{a}_B)$, determine the joint commands.

Step 3: Run the application. If the inequality expressed in (34) is not met and no means exists of improving calibration accuracy, the task cannot be performed. It should be pointed out that several important considerations arise in the second step just described. The inverse kinematic solutions derived with the assumptions of parallel or intersecting axes may no longer be valid for the “real” robot B. In this case either the solution to the inverse kinematics of the more general robot (robot B) configuration should be found numerically (see, for example, [20], [21] or a new closed-form solution which would represent a perturbation about the parallel or intersect-

ing axes solution should be developed. Determination of closed-form solutions of inverse kinematics of geometries that can be considered as small perturbation about more simplified geometries do not seem to have been investigated in the past.

The case of taught applications is even more involved than the data-driven mode. Let us consider a robot A that has been taught an application that involved a large number of task points. As a result of wear, part replacement, maintenance, robot replacement, or some other occurrence that alters the kinematic structure of the robot, a “new” robot B “appears” that now must perform the task of robot A. Presumably, it is undesirable to reteach the entire application. If the kinematic model of robot A is unknown, the most that can be done is to reteach robot B a subset of the task points and then to estimate the other task points through some form of interpolation. The feasibility of this technique is task-dependent. For example, the technique may be successful for tasks only using a small portion of the workspace. This approach, however, is applied at the joint level and does not fall under the definition of “robot calibration” used in this paper. Level 2 calibration, as defined in this paper, makes sense only when a nominal kinematic model is available.

If robot A underwent calibration prior to teaching the task, a more general type of correction becomes possible. If we assume that the information from the calibration of robot A is available and that the model of robot B has been identified, the goal is to find a transformation that maps the task points η_A into η_B so that the accuracy requirement (34) is met. This is expressed mathematically as follows:

$$\|g(\eta_B(j), \hat{\gamma}_B, \hat{a}_B) - g(\eta_A(j), \gamma_A, a_A)\| \leq \epsilon_{ac}(j). \quad (35)$$

In particular, the following additive correction is of interest:

$$\eta_B(j) = \eta_A(j) + \Delta\eta(j). \quad (36)$$

The requirement expressed in (35) may be approximated by

$$\|g(\eta_A(j), \hat{\gamma}_B, \hat{a}_B) + J\Delta\eta(j) - g(\eta_A(j), \gamma_A, a_A)\| \leq \epsilon_{ac} \quad (37)$$

where J represents a Jacobian matrix of robot B evaluated at the task points defined by robot A. Thus to a first-order approximation, the correction may be given by

$$\Delta\eta(j) = J^{-1}[g(\eta_A(j), \hat{\gamma}_B, \hat{a}_B) - g(\eta_A(j), \gamma_A, a_A)]. \quad (38)$$

In other words, the correction factor depends on the kinematic equations for robot A, the kinematic equations for robot B, and the Jacobian for robot B, all evaluated at the task points defined by robot A.

As is apparent from (38), the first-order approximation breaks down whenever a large variation occurs between models A and B or when a task point is mapped into the vicinity of a singular configuration of robot B. This raises an interesting issue involving workspace and task planning. If we are given a robot A with a known kinematic model as well as a statistical model of X (the kinematic offset vector), how may we determine the “available workspace” for taught applica-

tions so that the first-order correction will provide a suitable calibration? Two basic requirements must be met.

1) One should not teach points that may go outside the workspace of all possible “robots B”.

2) One should not teach points that may enter the “singularity regions” (see, for example, Paul and Stevenson [22]) of any possible robot B.

These workspace issues suggest further studies into the problem of predicting the effect of small perturbations of the kinematic parameters on robot singularities. For example, the relationship between the singular points of two similar robots has yet to be determined, although questions along these lines have been raised by certain investigators (see, for example, [23]).

LEVEL 3 CALIBRATION

As described earlier, level 3 calibration will be defined as “non-kinematic” calibration. The basic assumption underlying the level 2 calibration was that the position and orientation of the end effector could be defined as a function of only the joint displacements and the kinematic structure of the robot. This assumption is valid only if, for example, the links of the robot are perfectly rigid, the joints are frictionless and have no compliance, and the robot is not under dynamic control. The degree to which these assumptions are valid depends upon the robot, its motion control software, and the task it is to perform.

If it is determined that the model used in the level 2 calibration is not sufficiently accurate, or if the robot is under dynamic control, then one must consider some type of level 3 calibration. At this level, a number of effects exist which may be modeled. For example, deflection of the robot links or the effects of joint compliance and gear backlash may be significant. To include such factors, one must modify the basic form of the model to capture such effects within the position control software of the robot. In the work done by Whitney *et al.* [2] several aspects of joint compliance and gear backlash have been considered.

In cases where the robot is under dynamic control, the model used in the position control software takes the following form:

$$x = l(\eta, \gamma, a, \dot{x}, \ddot{x}, \mu) \quad (39)$$

where $\dot{x}$ and $\ddot{x}$ represent both the translational and angular velocity and acceleration of the end effector and μ represents the set of coefficients of the dynamic terms in the equations of motion for the robot. It must be emphasized here that (39) represents an extremely complicated functional relationship. Even if the links of the robot are assumed to be rigid and the joints perfect, the dynamic equations relating joint torques to the motion of the end effector can become quite involved (see Paul [24]). For this reason only a limited amount of work has been done in this area of level 3 calibration. Rather than trying to generalize such level 3 formulations, as was done for levels 1 and 2, we will only attempt to briefly review some of the relevant work in this area.

For the most part, existing work on level 3 calibration has

involved the identification of the mass and inertial parameters for the various links in the robot. For example, Neuman and Khosla [25] describe a technique for determining the dynamic parameters of a rigid-link robot having a known kinematic structure. Mukerjee and Ballard [18] also address this problem with the inclusion of a model of friction at each of the joints.

In general, many research issues need to be resolved before level 3 calibration can become widely used. For example, in calibrating under dynamic control, measurements of end-effector velocity and acceleration are required as well as position. This complicates the measurement phase considerably. The complexity of the dynamic model combined with the uncertainty of the data from the measurement make both the identification and correction phases challenging problems. Some work, however, has been done in this area in relation to identification of the dynamic model of a robot for control purposes (see, for example, [26]). A description of such works are, however, beyond the scope of this paper.

CONCLUSION

In relation to improving robot accuracy by calibration, we have reviewed the existing work and have identified several unresolved research issues. We have pointed out that improvement in robot accuracy, due to calibration, is limited by robot repeatability and the calibration accuracy. Calibration accuracy by itself depends on the accuracy of the measurement system and the inaccuracies due to robot parameters that are not modeled. We have defined three levels of calibration and have shown that level 1 and 2 calibrations are the most widely studied as compared to level 3 calibration.

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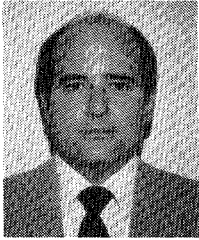
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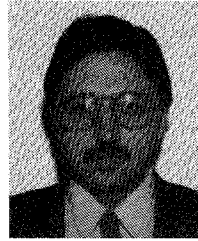
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